ARTOGRAM

WHERE ART MEETS IDENTITY

CORE PLATFORM DISTINCTION:
ARTOGRAM is intentionally built as an Instagram-model visual social network engineered specifically for artists, traditional artisans, and cultural lineages to showcase their creative talents, engage with followers, and build authentic audiences — *not just a generic administrative database, corporate dashboard, or static website.*

Architecture: React 19 + Django REST + PWA + SQLite/PostgreSQL	**Security & Integrity:** SHA-256 Byte Hashes + Physical QR Codes
Central Feature: Artogram Digital Identity Card Component	**Author:** Joseph K Joy (Lead Engineer & Architect)
Target Environment: Localhost (Fully Operational & Seeded)	**Repository:** github.com/joseph-k-joy/Artogram

1. The Problem Context

Why Traditional Artists and Cultural Creators Are Failed by Generic Social Media

1. Attribution Stripping When artworks, tribal motifs, and heritage designs are uploaded to mainstream social platforms (Instagram, Pinterest, X), metadata and context are instantly stripped. Reposts and scrapers erase original creator credits, leading to cultural erasure and theft.	**2. Unauthorised Commercial Scraping** Automated scrapers and e-commerce bot storefronts rip high-res art from public social feeds to sell molded replicas, apparel prints, and dropshipped goods with zero royalties, consent, or lineage recognition.
3. Broken Digital-Physical Bridge Physical exhibits in galleries, museum collections, and artisan craft fairs lack an authoritative, instant way for visitors to scan an artwork on a wall and inspect its verifiable provenance, creator declaration, and authentic background.	**4. Zero Legal & Covenant Clarity** Mainstream social networks provide all-or-nothing copyright terms. Artists have no standard mechanism to permit educational research while strictly banning unauthorized commercial exploitation or sacred motif desecration.

THE ARTOGRAM SOLUTION: Transforming the Instagram creator paradigm with an underlying cryptographic provenance engine, permanent sequential IDs, and verifiable digital Identity Cards.

2. The Instagram Model Transformed for Creators

A True Social Platform Where Artists Showcase Talents — Not Just A Generic Website

Mainstream Social Platform (Instagram)	ARTOGRAM Cultural Creator Platform
• Ephemeral, algorithmic consumption feed prioritizing paid ads	• **Visual Creator Timeline (`/feed`):** Pure focus on authentic artwork showcasing, creator portfolios, and cultural discovery
• Strips image metadata; encourages uncredited viral re-sharing	• **Attribution-Preserving Sharing:** Internal share modal transmits artwork identity, creator credits, and provenance card intact
• Likes and comments lack cultural context or creator covenants	• **Respectful Engagement Engine:** Live duplicate-safe likes, threaded comments, saved vaults, and follow graphs
• No physical world connection or verification link	• **Physical/Digital QR Bridge:** Every post connects to an immutable physical Identity Card and public verification certificate
• No formal reuse licensing or attribution dispute mechanism	• **Permission Hub & Misuse Reports:** Built-in workflows for academic reuse licensing and evidence-backed misuse reporting

3. Signature Innovation: The Artogram Identity Card

A Standardized, Shareable & Downloadable Digital Identity for Every Registered Work

Component: ArtworkIdentityCard.jsx	Key Functional Capabilities
1. Visual Artwork Showcase: High-resolution preview with lineage badge. **2. Attributed Creator & Origin:** Explicit creator name, username, and cultural region. **3. Sequential Provenance ID:** Human-readable identifier (e.g. ART-2026-000001). **4. Cryptographic Checksum:** 64-char SHA-256 fingerprint computed from raw bytes. **5. Verifiable QR Matrix:** Resolves directly to public provenance certificates. **6. Reuse Covenants Matrix:** Clear permissions for educational vs. commercial use. **7. Mandatory Credit Notice:** Standardized citation string for researchers.	• **1-Click High-Res PNG Export:** Integrated with html2canvas to render and export crisp 300 DPI graphics directly in the browser. Galleries, museums, and craft stalls can print this card for physical exhibition display. • **Zero-Login Public Verification:** Scanning the card's QR code on any smartphone camera immediately opens the public certificate route /artworks/ART-YYYY-XXXXXX/provenance/ with full audit events. • **Social Transmission:** Users can share the card internally with other members or copy direct public verification URLs.

ARCHITECTURAL SEPARATION OF CONCERNS:
Artwork (Art asset & metadata) | ArtworkProvenance (SHA-256 & QR) | SocialPost (Feed presentation) | ArtworkIdentityCard (Exportable component).
An artwork can be featured in multiple social posts without minting duplicate provenance records.

4. System Architecture & Technology Stack

Enterprise-Grade Full-Stack Implementation Built for Robustness & PWA Installability

Frontend (React 19 & Vite 8)	Backend (Django 6 & DRF 3.18)	Persistence & Operations
• **React 19 & Vite 8:** Blazing fast SPA architecture. • **Tailwind CSS v4:** Custom earthy cultural palette. • **React Router v7:** Client-side routing with role guards. • **Axios + JWT Interceptors:** Automatic token refresh. • **PWA Plugin:** Precaching service worker & manifest. • **html2canvas:** Client-side PNG card rendering. • **Lucide Icons:** Accessible visual iconography.	• **Django 6.0 & DRF 3.18:** Modular app architecture. • **Simple JWT 5.5:** Access & refresh token rotation. • **Python hashlib:** SHA-256 raw byte fingerprinting. • **Python qrcode & Pillow:** Physical QR image matrices. • **Append-Only Audit Engine:** Immutable event recording. • **Granular RBAC:** Visitor, User, Artist, Moderator, Admin. • **Seed Data Engine:** Realistic cultural showcase data.	• **PostgreSQL 16:** Production-ready database container via Docker Compose. • **SQLite 3 Fallback:** Instant zero-config local run. • **Django Media Storage:** Local disk media with Cloudinary adapter support. • **Automated Testing:** 17 unit and integration tests covering all critical workflows. • **Localhost Ready:** 100% runnable without third-party cloud costs.

PWA READINESS: Artogram installs as a standalone native app on Android, iOS, Windows, and macOS, featuring an offline fallback page (/offline.html), precached assets, and automatic install prompt handling.

5. Ethical Governance, Permissions & Integrity

Bridging Contemporary Art Sharing with Cultural Sovereignty & Accountability

Reuse Permission Hub	Misuse Reporting Pipeline	Community Guild Verifications
Institutions, museums, and educators can formally request reuse licenses. Artists approve or reject requests with custom covenants, automatically generating notifications and recording decisions in the audit log.	Users can report suspect misuse (missing attribution, unauthorized sales, sacrilege). Transparent lifecycle: SUBMITTED → UNDER REVIEW → ACTION TAKEN / RESOLVED / REJECTED. Evidence URLs & screenshots inspected by admins.	Designated cultural councils (e.g. *Sahyadri Warli Artisan Council*, *Egba Adire Guild*) review traditional claims. Elevates artworks to **Community Verified & Approved**, clearly distinct from self-registered works.

ADMIN CONSOLE (`/admin/dashboard`): Features live 8-metric telemetry, user role promotion, comment moderation (hide/unhide), report review, and an append-only database audit history.

6. Limitations & Future Development Scope

Transparent Engineering Bounds & Forward-Looking Research Trajectory

Current Engineering & Legal Boundaries	Future Development Roadmap
1. Cryptographic Scope of SHA-256: SHA-256 certifies exact byte-level integrity. It proves the exact file has not been altered, but cannot detect cropped, resized, or filter-modified copies. **2. Attribution Claims vs. Legal Copyright:** Artogram records cultural provenance and creator declarations; it does not replace government copyright registries. **3. Localhost Demonstration Scope:** In hackathon mode, files are stored on local Django media and tested on local SQLite/Postgres.	**1. Neural Perceptual Hashing (pHash + AI):** Combine exact SHA-256 with perceptual neural embeddings to identify cropped or color-shifted stolen art copies across the web. **2. Automated Misuse Web Crawler:** Background daemon periodically querying public search engines and social platforms to alert artists if their art is re-uploaded uncredited. **3. Multi-Signature Tribal Elder Consensus:** Require multiple verified trustees to sign off on community-held heritage claims. **4. Decentralized Storage Anchoring:** Optional IPFS/Arweave permanent decentralized storage mirroring.

7. Summary & Live Demo Credentials

Ready for Live Hackathon Evaluation & Localhost Demonstration

Role	Username	Password	Key Demo Journey To Present
Traditional Artist	maya_warli	ArtistPass2026!	Register cultural artwork, mint SHA-256 provenance & download Identity Card PNG
Cultural Patron	catherine_curator	UserPass2026!	Like & comment on feed, bookmark to vault, request educational reuse license
Administrator	artogram_admin	AdminArtogram2026!	Open /admin/dashboard, review misuse reports, approve verifications & inspect audit trail

FINAL TAKEAWAY:
ARTOGRAM proves that the modern, engaging visual social media model (Instagram) can be repurposed to protect creators instead of exploiting them. By giving every creative piece an immutable digital identity, verifiable QR code, and exportable card, Artogram delivers where art truly meets identity.